

Slide	Notes
C O D E Lesson 3: The AI's Brain Problem Solving with AI	
Warm Up	Warm Up (10 mins)
Problem Solving with AIThe AI's Brain – Warm Up Discuss: What would happen if you read every message ever sent in a giant group chat? Think about common phrases, emojis, or inside jokes you might notice. Fam, I'm exhausted today 🙄 Someone, I woke up way too late Late? Bro, I barely survived IT, who let Sunday disappear like that?? I did and it was Monday 🤔 And now we're back to the struggle The struggle is But you'll still be up till 2AM 🕒	■ Goal of this activity: To help students understand how patterns emerge from large sets of data and how those patterns influence predictions, similar to how neural networks function. ■■ Say: Imagine being in the world's biggest group chat. If you had access to every message ever sent, what might you notice? What would happen if you read them all? ■ Discuss: What would happen if you read every message ever sent in a giant group chat? ■ Discussion Goal: Students recognize that they would start noticing patterns, commonly used emojis, inside jokes, tone, and conversational flow. This sets up the idea that prediction is based on what comes before—just like how AI models work.
Problem Solving with AIThe AI's Brain – Warm Up Discuss: What patterns might you pick up over time? Could you start predicting what people might say next? How? How might your responses be shaped by what you've read, rather than your own original thoughts?	■ Discuss: Click through the animated slide to display the prompts. What patterns might you pick up over time? Could you start predicting what people might say next? How? How might your responses be shaped by what you've read, rather than your own original thoughts? ■ Discussion Goal: Students should recognize how behavior can become patterned by what they've read—just like AI does. Reinforce that: - AI models also read (or are trained on) millions of examples. - AI doesn't "think," it predicts based on pattern recognition. - The more data it has, the better it becomes at mimicking human-like responses. ■■ Say: Now that we've seen how AI can predict responses based on patterns in a group chat, let's take that concept further and explore how AI models, like neural networks, make predictions with limited context and how providing more context can help improve the accuracy of those predictions.

Lesson Objectives

By the end of this lesson, you will be able to . . .

- Explain how AI models, especially LLMs, use neural networks and training data.
- Analyze how training data shapes AI behavior and biases.
- Experiment with AI responses to identify learning patterns.
- Evaluate how AI's learning affects accuracy and reliability.

■ ■ Do This: Introduce the lesson objectives.

Question of the Day

How do AI models learn?

■ ■ Do This: Introduce the Question of the Day.

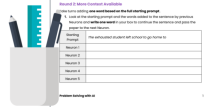
Activity

Activity (30 mins)

Unplugged Neural Network

Each group should have:

- Unplugged Neural Networks activity guide
- pen or pencil



Unplugged Neural Network (10 mins)

■ Goal of this activity: To demonstrate how neural networks predict text based on limited context and show how increasing context improves the accuracy of predictions.

■ ■ Do This: Place students in groups of three to five.

■ Distribute: Give each group one copy of the Unplugged Neural Networks activity guide, placed face down. Let students know they'll flip it over when instructed.

Unplugged Neural Network

Goal: Build a sentence based on . . .

- **Round 1** - the previous three words
- **Round 2** - the previous sentence



■ ■ Do This: Introduce the activity. Make sure students understand that the purpose is to simulate how an AI model makes predictions with limited and full context.

■ ■ Say: In this activity, your group will simulate how an AI model predicts words. In Round 1, you'll only get to see the last three words of a sentence before predicting the next word. That's how older models worked—with very limited memory. In Round 2, you'll get to see the entire previous sentence before making your prediction—like modern transformer models, which work with much more context. Let's see how that changes your predictions.

Do This:

1. **Neuron 1:** Flip over the paper, read the starting prompt, and add one word to continue the sentence.
2. Fold the paper so that only the last three words are visible. Pass it to the next person.
3. **Neurons 2–5:** Look at the three visible words, add one word to continue the sentence, then fold the paper again so only the last three words are showing. Pass it along.

■ **Do This:** Direct students to complete Round 1 in their activity guide. Use the steps on the slide to explain the activity procedure and answer any questions. When everyone has added a word, instruct student to unfold the paper and read the full sentence aloud as a group. Have groups to compare their sentences in order to see how similar or different they were.

For example, students' sentences could end up like this:

Space travel is
exciting
and
dangerous
animals
live.

■ **Circulate:** Circulate to ensure students aren't peeking back at earlier words or the full sentence.

Do This:

This time, everyone can see the full sentence and all added words.

1. **Neuron 1:** Add a word to the sentence.
2. Pass the paper (**unfolded**) to the next neuron.
3. Repeat until each group member has added a word.

■ **Do This:** Direct students to complete Round 2 in their activity guide. Use the steps on the slide to explain the activity procedure and answer any questions. When everyone has contributed, read the full sentence aloud as a group. Direct groups to compare their sentences to Round 1—how did having more context change the predictions?

Example:

The exhausted student left school to go home to . . . rest before starting homework.

The exhausted student left school to go home to . . . eat dinner with the family.

■ **Circulate:** As students read full sentences, listen for students noticing how more words made it easier to choose a logical next word. If students aren't making the connection, ask: "What helped you decide which word to add this time?" and "Was it easier or harder than the first round?"

Teaching Tip

Due to the variety of answers possible, there is no exemplar provided for this activity. Reference the examples above for possible student generated sentences.

Discuss:

How did having limited context affect predictions?

What changed when you had access to the entire starting sentence and all words that were added before you?

■ **Discuss:** Click through the animated slide to display the prompts.

How did having limited context affect predictions?

What changed when you had access to the entire starting sentence and all words that were added before you?

■ **Discussion Goal:** Students recognize that limited context leads to less accurate and more random predictions, while access to more context helps predictions become more logical and connected to previous content. This mirrors how modern AI, like transformers, improves its output by analyzing longer sequences of data.

Discuss:

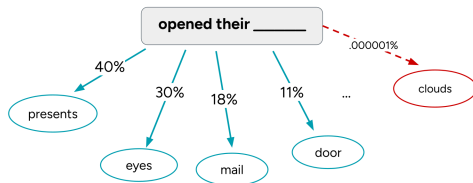
In the neural network activity, predictions improved when more context was provided.

How do you think this applies to AI prompts?

■ **Discuss:** In the neural network activity, predictions improved when more context was provided. How do you think this applies to AI prompts?

■ **Discussion Goal:** Students recognize that just like their own predictions, AI performs better when it has more context. The more details and structure we give in a prompt, the more accurate, relevant, and helpful the AI's response becomes. Encourage students to connect this back to their experience seeing the full sentence in Round 2.

AI Predictions

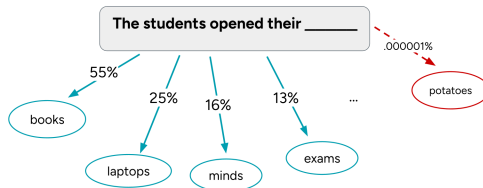


■ Do This: Have students look at this slide. Explain that generative AI uses probabilities to predict what word should come next in a sentence. In this example, AI is most likely to predict the word "presents" or "eyes" next. AI is extremely unlikely to choose the word "clouds", which is why the percentage is so small. Since this model has limited context, i.e. it only sees the words "opened their", sometimes its predictions don't make sense in the context of what came before it.

Teaching Tip

Make sure students understand that the percentages represent how likely AI is to choose that word to come next. In this example, AI is most likely to choose the word "presents" (40%) next. It is slightly less likely to choose "eyes" (30%) and much less likely to choose "mail" (18%), "door" (11%), or any other word.

Improved AI Predictions



■ Do This: Have students look at this slide and note the differences between this slide and the previous slide.

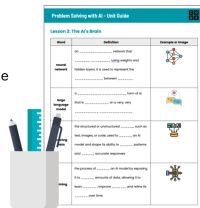
■ Discuss: Why did AI's predictions change? How does this relate to the neuron activity?

■ Discussion Goal: Students should be able to explain that increased context allows AI to make more appropriate predictions. Here, the inclusion of the words "The students" led the AI to make next word predictions related to a school setting.

Unit Guide

You should have:

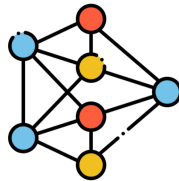
- Problem Solving with AI Unit Guide
- pen or pencil



■ Do This: Have students take out their Problem Solving with AI Unit Guide.

What is a neural network?

A **neural network** is an interconnected network that makes decisions using weights and hidden layers. It is used to represent the relationship between words.



Unit Guide

■ Do This: Define neural network.

Video: Chatbots and Large Language Models

What is the difference between LLMs and typical neural networks?



How LLMs Learn (10 mins)

■ Goal of this activity: To help students understand how large language models (LLMs) like ChatGPT learn from data, recognize patterns, and refine responses through human feedback.

■ Video: Show the "Chatbots and Large Language Models" video (0:00-1:20).

Questions to consider with the video:

What is the difference between LLMs and typical neural networks?

Discussion Goal: Students recognize that typical neural networks perform specific tasks, while LLMs process vast text data to generate responses. LLMs predict words using probabilities, making them more flexible and capable of producing human-like text.

What is a Large Language Model?

A **large language model (LLM)** is a general purpose form of AI that is trained on a very, very large body of knowledge



Unit Guide

■ Do This: Define large language model.

Video: Chatbots and Large Language Models

- Why is it important for the AI to look at longer sequences such as sentences or paragraphs?
- How does this relate to the context we provide in our prompts?



■ Video: Show the "Chatbots and Large Language Models" video (1:20-1:51).

Questions to consider with the video:

Why is it important for the AI to look at longer sequences such as sentences or paragraphs?

How does this relate to the context we provide in our prompts?

Discussion Goal: Students recognize that longer sequences provide more context, leading to more accurate predictions. They connect this to AI prompting, understanding that detailed prompts help AI generate more relevant and coherent responses.

Video: Chatbots and Large Language Models

- How does human feedback improve AI responses?
- Why is it still important to be cautious when using AI, even with fine-tuning?



■ Video: Show the "Chatbots and Large Language Models" video (1:51-2:30).

Questions to consider with the video:

How does human feedback improve AI responses?

Why is it still important to be cautious when using AI, even with fine-tuning?

Discussion Goal: Students recognize that human feedback helps AI refine responses by correcting errors and reducing biases. They also understand that, despite fine-tuning, AI can still produce inaccurate or biased outputs, requiring users to think critically about its responses.

What is training data?

Training data is the structured or unstructured information, such as text, images, or code, used to train an AI model and shape its ability to recognize patterns and generate accurate responses.

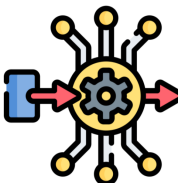


Unit Guide

■ Do This: Define training data.

What is training?

Training is the process of teaching an AI model by exposing it to large amounts of data, allowing it to learn patterns, improve accuracy, and refine its predictions over time.



Unit Guide

■ Do This: Define training.

Prompting Challenge

Lesson 3

1 ○ ○ ○ ○ ○

Do This:

1. Go to Lesson 3, Level 1.
2. Choose a career path and practice refining your prompts on Levels 1 and 2.
3. Record your answers in the Unit Guide after each level.

[Unit Guide](#)


Prompting Challenge (10 mins)

■ **Goal of this activity:** To give students hands-on practice with AI prompting, focusing on how context, examples, and structured inputs improve AI-generated responses. Students will explore real-world career scenarios and experiment with zero-shot, one-shot, and multi-shot prompting to refine AI responses.

■■ **Say:** You've seen how AI predictions get better when given more context. Now, you're going to test that idea by giving the AI different types of prompts—starting with no example, then adding one, and finally, adding multiple examples. Try to notice how the responses change at each level.

■■ **Do This:** Pair students up.

■ **Transition:** Direct students to Lesson 3, Level 1 on Code Studio to complete Levels 1 through 3 of the Prompting Challenges activity. For each level, students will choose one of three options based on a real-world career path:

Level 1: Students provide a basic, unstructured prompt for their chosen career path.

Level 2: Students add one example to guide the AI's response.

Level 3: Students refine the prompt with multiple examples and structure to improve accuracy.

■ **Circulate:** Ensure students are working collaboratively with their partners and completing the Prompting Challenges section of their Unit Guide after each level, noting how the AI's responses changed and what role context played in improving them. Listen for how they describe the differences between each AI response as they add more context. Encourage students to be specific in their reflections. If students are unsure what to write, ask:

What changed in the AI's response when you gave an example?

Did the response get more accurate, creative, or specific? How do you know?

What kind of details made the biggest difference?

Teaching Tip

Each level includes multiple career path options. If time allows, encourage students to try more than one scenario to see how prompting techniques apply across different contexts. Ask them whether their observations about AI responses stay consistent across tasks or change depending on the topic.

Assessment Opportunity

Use student reflections in the Unit Guide to assess how well students understand the role of context and structure in AI responses. Look at their answers to the reflection questions after each level (zero-shot, one-shot, and multi-shot). Based on their responses, determine next steps:

If students give vague answers or say AI's response didn't change much across levels:

Reteach the importance of context by reviewing the neural network activity analogy. Remind students that AI improves with more context and clearer instructions.

Prompt with a discussion question: "Why would AI struggle to give useful results if your prompt is too short or too open-ended?"

Small group discussion: Have students compare responses at different levels and identify specific changes in tone, structure, or accuracy.

If students show misunderstanding (e.g., they say multi-shot was worse or don't reference how examples helped):

Review sample prompt chains together as a class. Ask: "What kind of details helped this version get better?"

Have students rewrite one of their weaker prompts from earlier levels using stronger structure or clarity and re-try it with the AI.

If students clearly articulate how adding structure and examples improved the AI's response:

Move on without reteaching. These students demonstrate understanding of prompt engineering basics.

Optional Challenge: Encourage them to try a second career scenario and see if their insights apply across contexts.

Discuss:

What's a creative use of this prompting technique?

■ Discuss: What's a creative use of this prompting technique?

■ Discussion Goal: Students share creative ways one-shot and multi-shot prompting could be used, giving a quick snapshot of how students envision AI supporting creativity or problem-solving.

Teaching Tip

Keep it brief with popcorn-style sharing (3–4 ideas max) and note emerging themes to revisit later when discussing real-world or ethical uses of AI.

Wrap Up

Wrap Up (5 mins)

Today, you learned about . . .

- how AI models, especially LLMs, use neural networks and training data.
- how training data shapes AI behavior and biases.
- experimenting with AI responses to identify learning patterns.
- how AI's learning affects accuracy and reliability.

■ Do This: Review the lesson objectives.

Question of the Day

How do AI models learn?

Question of the Day Comic Strip (5 mins)

■ Do This: Review the Question of the Day and give students a moment to just think about what they learned today. Let students know that they will be answering the question a little differently today.

Unit Guide

Do This:

Create a short comic strip that shows something you learned about how AI learns, predicts, or responds to prompts—make it funny, insightful, or even a little weird!



Unit Guide

■ Do This: Direct students to create a short comic strip in their Unit Guide that illustrates something they learned about how AI learns, predicts, or responds to prompts, encouraging humor or creativity. Let students know that this is a creative way to synthesize the lesson. They can be funny, serious, or weird—as long as it's related to how AI works.

■ Circulate: Look for students who incorporate ideas like limited context, prediction, pattern recognition, or prompting. If students are stuck, suggest they focus on one concept (e.g., how AI uses training data, how it predicts the next word, or how adding more context makes its output better).



Key Vocabulary

- **neural network:** an interconnected network that makes decisions using weights and hidden layers. It is used to represent the relationship between words.
- **large language model (LLM):** a general purpose form of AI that is trained on a very, very large body of knowledge



■ ■ Do This: Review the Key Vocabulary.



Key Vocabulary

- **training data:** the structured or unstructured information, such as text, images, or code, used to train an AI model and shape its ability to recognize patterns and generate accurate responses
- **training:** the process of teaching an AI model by exposing it to large amounts of data, allowing it to learn patterns, improve accuracy, and refine its predictions over time



■ ■ Do This: Review the Key Vocabulary.